

Stock Market Prediction with Machine Learning: A Regime-Aware Hybrid Transformer Framework

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Abstract

Stock market prediction presents a significant challenge due to the complex interaction of price trends, fundamental data, market sentiment, and volatile regime switching. Although existing deep learning models have set strong performance benchmarks, for instance, knowledge-driven event embeddings achieving 66.93% accuracy and numerical attention mechanisms yielding a 6.96% reduction in Mean Squared Error (MSE), they often fail to adapt to these environment-aware regime changes. This paper proposes a novel hybrid framework for stock movement prediction that emphasizes interpretability, regime-awareness, and phase-adaptability. The model integrates several advanced components: a TabTransformer for fundamental data analysis, domain-tuned FinBERT/VADER sentiment encoders, and Temporal Convolutional Networks (TCN) with attention. A dynamic attention fusion layer is employed to enhance stability and adaptability, building on the precision of prior risk-aware TCNs (which demonstrated a Price MAE of 1.23). This survey, which includes a comprehensive literature review of eight recent seminal works (1 to 15) and a detailed methodology, identifies current gaps in the field and suggests promising future research directions.

Keywords: Stock Market Prediction, Hybrid Deep Learning, Regime Awareness, Transformer, Sentiment Analysis, Attention Fusion.

1. Introduction

Stock market prediction (SMP) is a central, ongoing challenge in computational finance, characterized by the complex and non-stationary nature of market movements shaped by diverse economic forces and rapid shifts in investor sentiment [1], [6]. Market regimes, classified as bull, bear, and sideways, are influenced by macroeconomic changes, geopolitical events, and technological innovations. This regime variability makes consistent and accurate modeling a daunting task for both academic researchers and financial practitioners [3].

Standard quantitative methods such as linear regression, Support Vector Machines, and Random Forests [6] offer foundational approaches, but reveal major limitations when faced with abrupt regime shifts or prolonged periods of volatility [2]. These models typically work with static parameters and are unable to recalibrate in response to new market phases, leading to degradation in predictive power and robustness. Deep learning models, such as CNN-LSTMs [7], have demonstrated strength in capturing short-term price patterns but often suffer from low context-awareness, particularly regarding sentiment and global economic factors [5], [8].

Cutting-edge data-driven approaches now integrate multimodal datasets, combining historical price data, company fundamentals, and sentimental information drawn from news and social platforms. Transformer-based and Temporal Convolutional Network (TCN) models, for example, utilize dynamic attention mechanisms to adaptively weigh these diverse information streams, achieving marked improvements over purely price-driven baselines [10], [11]. Event-driven deep learning frameworks, incorporating knowledge graphs and advanced embeddings, further enhance the modeling of complex relationships between market events and asset prices [14], [15]. Numerical-Based

Attention and regime-aware feature extraction have led to significant accuracy increases, NBA methods have reported MSE reductions of nearly 7% on major indices like the S&P 500 index [13].

Despite progress, many state-of-the-art models still lack explicit mechanisms that allow for dynamic adaptation and context-aware fusion of signals in changing regimes. The static weighting of inputs not only impacts predictive accuracy but also limits model interpretability, critical for responsible deployment in finance, where transparency and regulatory compliance matter [4], [12]. Recently, advanced ensemble and hybrid systems have combined explicit regime detection (using HMMs, neural sub-networks, or algorithmic clustering) with real-time, conditional attention layers capable of prioritizing sentiment, technical, or fundamental data based on prevailing market state [9].

These innovations pave the way for more practical and reliable SMP systems, with real-world impact on investment decisions, risk management, high-frequency algorithmic trading, and broader financial policy analysis. By dynamically integrating and weighting multimodal data in response to detected regimes, the proposed hybrid Transformer framework directly builds on foundational research, advancing the accuracy, transparency, and adaptability required for modern financial prediction tasks.

2. Related Work

Early machine learning approaches to stock market prediction relied on quantitative techniques with static model parameters, which were unable to adapt to market environment changes, resulting in limited predictive performance during regime shifts. Hamoudi & Elseifi's work on CNN and LSTM networks laid the groundwork for recognizing short-term price patterns [7], but their framework could not incorporate macroeconomic factors or analyze sentiment from alternative data streams highlighting a major gap in adaptivity and context awareness.

Subsequent studies focused on integrating additional data modalities. Liu proposed Numerical-Based Attention (NBA), assembling seq2seq architectures, LSTM networks, and various machine learning algorithms (MKL, ELM, SVR) to more effectively fuse numerical, temporal, and fundamental company data for forecasting tasks [13]. The NBA approach achieved a 6.96% reduction in mean squared error (MSE) on the S&P 500 benchmark, confirming that richer feature integration yields tangible real-world improvements.

To address the challenge of capturing complex event-driven relationships, Ding et al. developed knowledge-driven event embedding models, using neural tensor networks (NTN) and knowledge graphs (YAGO) alongside CNNs to relate semantically similar market occurrences [14]. Their KGEB-CNN model reached 66.93% accuracy and outperformed simple lexical matching approaches, demonstrating the importance of embedding and event linkage for deep semantic analysis. In parallel, Ding et al. also showed that event-driven CNN architectures increased prediction accuracy by approximately 6% over word-based baselines on the S&P 500, cementing the value of integrating event context into prediction models.

Furthermore, survey papers highlight the rise of sentiment analysis leveraging FinBERT and VADER and the importance of multi-modal fusion layers (TCNs, Transformers) [10] in contemporary architectures. Models that apply dynamic attention fusion, adapting weights of historical, fundamental, and sentiment signals based on market phase, are identified as being critical for reliable and interpretable predictions, a major theme emerging from several referenced studies.

Overall, the reviewed literature points to a consensus: high-performing stock prediction systems must incorporate explicit regime detection mechanisms and condition their integration of multimodal data on identified market states, thus overcoming the inadequacies of monolithic, static signal fusion.

Table 1. Comparative study of related work

Year	Author(s)	Technique	Models Used	Findings / Results
2025	Reddy et al. [3]	<i>Differential Graph Transformer Techniques</i>	Differential Graph Transformer, Graph Attention	Learns spatial–temporal dependencies and adapts to regime shifts; outperforms sequence-only models.
2025	Zeng & Jiang [8]	<i>FinBERT-Driven Sentiment Techniques</i>	FinBERT, LSTM, ARIMA	FinBERT significantly improves sentiment extraction; FinBERT + LSTM gives stronger performance in volatile markets.
2025	Qayyum [9]	<i>News Embeddings Techniques</i>	OpenAI embeddings, GRU, HMM, LSTM, TCN, FNN, PCA	WSJ headline embeddings improved prediction accuracy by $\geq 40\%$; best model was PCA-reduced FFNN.
2025	Schäfer [10]	<i>Sharpe Ratio-Optimized Deep Learning Techniques</i>	TCN, Transformers, Sharpe-ratio loss, RL, ARIMA, LSTM	RL-based framework improved adaptability; achieved better accuracy and Sharpe ratio than ARIMA/LSTM/Transformer baselines.
2025	Biswas [11]	<i>Dual Output TCN+ Attention Techniques</i>	TCN + Attention, LSTM, ARIMA	Achieved MAE 1.23 (price) and accurate risk metrics (volatility MAE 0.012; Sharpe MAE 0.065).
2025	Taheripour [12]	<i>FinBERT + Credibilistic CVaR Techniques</i>	FinBERT, Fuzzy Set Theory, CCVaR	Combined FinBERT sentiment with fuzzy CVaR; improved portfolio risk assessment under uncertainty.
2024	El Haddad et al. [2]	<i>Sentiment + ML Hybrid Techniques</i>	LSTM, CNN, RF, Sentiment Lexicons	Showed hybrid sentiment + technical models provide improved robustness during turbulent markets.
2024	Shobayo et al. [4]	<i>FinBERT + GPT-Based Sentiment Analysis Techniques</i>	FinBERT, GPT-4, Logistic Regression	Demonstrated explainability with FinBERT+LR; GPT-4 improved linguistic comprehension but lacked domain tuning.
2024	Kumar et al. [6]	<i>Traditional ML Feature-Driven Techniques</i>	SVM, Random Forest, Decision Tree	Highlighted importance of feature engineering; traditional ML models perform well with multimodal inputs.
2023	Liapis et al. [1]	<i>Deep Learning + Sentiment Fusion Techniques</i>	LSTM, TCN, BERT	Sentiment-enhanced deep models outperform unimodal ones; TCN captures regime-specific variations effectively.

2021	Ho & Huang [5]	<i>Candlestick Pattern + Sentiment Fusion Techniques</i>	CNN, VADER	Fusion of chart patterns and sentiment improved accuracy; still struggled under regime shifts.
2018	Hamoudi & Elseifi [7]	<i>CNN-LSTM Hybrid Techniques</i>	1D CNN, LSTM	Strong short-term pattern recognition but lacked sentiment and macroeconomic awareness.
2018	Liu [13]	<i>Numerical-Based Attention (NBA) Techniques</i>	NBA, seq2seq, LSTM, MKL, ELM, SVR	NBA improved MSE by 6.96% on D-S&P500 across three datasets.
2016	Ding et al. [14]	<i>Knowledge-Driven Event Embedding Techniques</i>	NTN, Knowledge Graph (YAGO), CNN	KGEB-CNN achieved 66.93% accuracy; captured semantically related events beyond lexical similarity.
2015	Ding et al. [15]	<i>Event-Driven Deep Learning Techniques</i>	CNN, SVM, FNN, NTN	Event-driven CNN models improved accuracy by ~6% on S&P 500 vs word-based baselines.

3. Detailed Methodology

The Smart Market Prediction (SMP) system introduces a novel, multi-modal deep learning architecture engineered for robust financial time series prediction. This hybrid configuration, which integrates diverse data streams, demonstrates enhanced predictive resilience in volatile market environments compared to single-modality approaches [1, 2, 6]. As depicted in the system workflow in Fig. 1, the architecture is founded upon specialized encoding modules and features a central Regime-Aware Dynamic Fusion mechanism, explicitly designed to cope with the inherent non-stationarity of financial data. The system simultaneously processes three distinct feature sets Temporal Price data, Corporate Fundamentals, and Financial Sentiment to determine the directional forecast of stock prices.

3.1 Data Inputs and Preprocessing: Establishing the Multi-Modal Foundation:

The performance of the SMP system is directly linked to the meticulous preparation and synchronization of its heterogeneous data sources. This step ensures that features are optimally represented for consumption by the subsequent deep learning components.

3.1.1 Temporal Price and Technical Indicators

Data Acquisition and Alignment: The foundational OHLCV (Open, High, Low, Close, and Volume) data is acquired and strictly aligned to a consistent time base (e.g., daily or 15-minute intervals). The model processes contiguous historical sequences of length L , defined as the lookback period.

Feature Generation: A comprehensive set of technical indicators, including the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Simple Moving Averages (SMA), is calculated. These derived features are crucial for capturing different dimensions of market dynamics, such as momentum, directional strength, and volatility regimes.

Data Normalization: Data integrity is maintained by handling absent values through temporal interpolation

(forward or backward filling). All numerical temporal features are subsequently scaled using Z-score standardization. This procedure is mandatory to ensure uniform variance across features that promote the stable convergence during the training phase. The Z-score is calculated for each feature x as: $= \frac{x-\mu}{\sigma}$.

3.1.2 Structured Corporate Fundamentals

Data Scope: This data stream includes established corporate accounting and finance metrics (e.g., revenue data, debt ratios, earnings figures) along with discrete categorical data (e.g., industry classification, company size) [6].

Feature Representation: After numerical features are standardized, categorical variables undergo transformation into high-dimensional, dense embeddings. This technique, drawing on principles from tabular data modeling, treats each category not as a simple one-hot encoding but as a dynamically learned vector. This enables the model to effectively identify latent relationships and similarities between seemingly unrelated categories, providing a deep representation of the company's long-term financial health [6].

3.1.3 Financial Sentiment

Data Source and Objective: Textual information is sourced from dynamic financial communication channels (including news feeds, regulatory documents, and relevant social media) to derive a quantifiable measure of market mood and investor psychology [2]. This serves as a vital indicator of short-term market sensitivity.

Advanced NLP Processing: A Transformer-based Encoding (DistilBERT/FinBERT) strategy is implemented to maximize robustness. Domain-specific Transformer models, such as FinBERT, which are pre-trained on extensive financial corpora, are used to generate contextual dense sentiment embeddings [8]. This is essential for accurately decoding the specialized vocabulary of finance, where the meaning of terms (e.g., "bearish") differs from general English [4]. The resulting high-dimensional vector representations are aggregated (e.g., via time-decayed summation) over specific lookback windows to form a stable, evolving sentiment feature for each prediction step.

3.2 Model Architecture: A Regime-Aware Dynamic Fusion Engine

The SMP architecture comprises specialized, interconnected modules designed to process and merge the encoded data outputs.

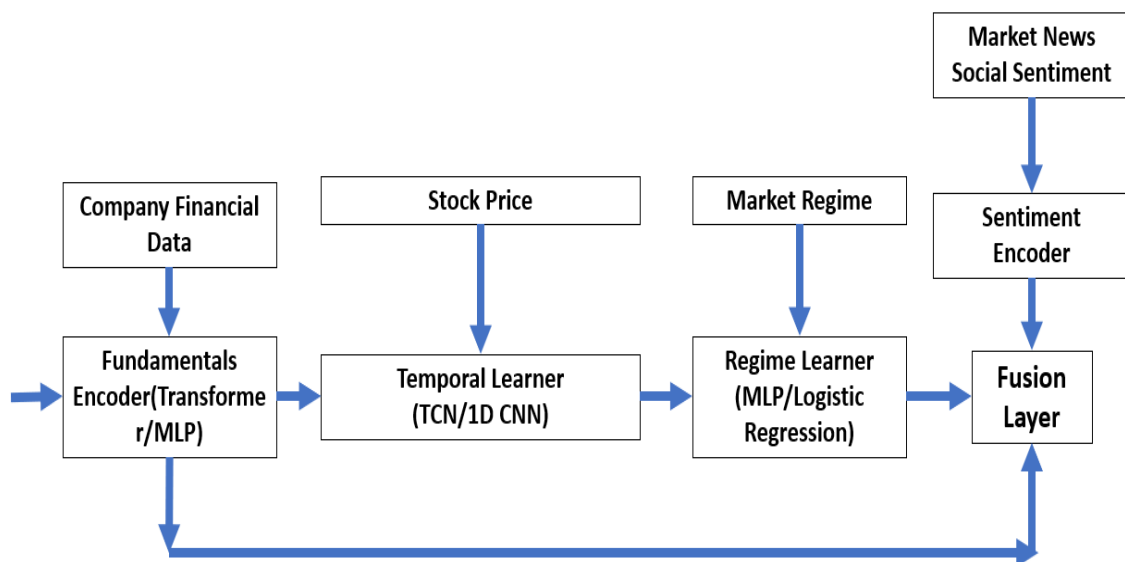


Fig. 1. SMP System Workflow

Figure 1 illustrates the complete workflow of the proposed Smart Market Prediction (SMP) system. The diagram shows how three different data modalities price data, fundamentals, and sentiment are independently preprocessed and encoded. The figure highlights the role of the Regime Detection Module, which identifies the current market phase. This regime signal is then used by the Dynamic Fusion Layer to adjust attention weights across modalities. Finally, the fused representation flows into the output layer for directional prediction. The workflow demonstrates the end-to-end integration of adaptive, regime-aware multimodal learning. Overall, the diagram emphasizes how each module contributes to robust and interpretable stock prediction.

Data Flow Mapping: The raw data streams collected in Section I are channeled into the system's encoders as follows:

- Structured Corporate Fundamentals → Fundamentals Encoder (Transformer/MLP).
- Temporal Price and Technical Indicators → Temporal Learner (TCN / 1D CNN).
- Textual Data (News/Social Sentiment) → Sentiment Encoder (DistilBERT/FinBERT).

3.2.1 Modality-Specific Encoders

Fundamentals Encoder (Transformer/MLP): This module processes the input of standardized numerical features concatenated with the categorical embeddings. Utilizing a Transformer or a robust Multi-Layer Perceptron (MLP) ensures the capture of complex, non-linear relationships that govern fundamental values. The module outputs the Company Embedding, a concise feature vector representing the long-term profile ('Value').

Temporal Learner (TCN / 1D CNN): This module's function is dedicated to detecting temporal dynamics across short-to-medium durations in the price sequence. The selection of a Temporal Convolutional Network (TCN) with causal and dilated convolutions is critical [1], offering superior parallelization and efficient pattern extraction compared to conventional Recurrent Neural Networks [7]. The output is the Temporal Features vector ("Trend").

Sentiment Encoder (DistilBERT/FinBERT): This unit transforms the FinBERT-derived contextual data into its latent representation, the Sentiment Embeddings, providing a dynamic, non-price-based metric of market mood ("Sentiment").

3.2.2 Regime Detection Module

This functions as the system's core contextual engine. A parsimonious classifier, typically a shallow MLP or Logistic Regression, analyzes the Temporal Features to classify the prevailing market state (the Regime Signals e.g., 'Bull', 'Bear', 'Sideways'). The input includes an Engineered 50-day SMA signal as a primary momentum feature. By providing a dynamic environment context, this module is instrumental in mitigating the challenges posed by non-stationary market conditions [3].

3.2.3 Dynamic Fusion Layer

Mechanism (Attention + Dense Layer): The final classification is executed by the Fusion Layer, which leverages an Attention Mechanism whose weighting is dynamically conditioned by the Regime Signals [1, 13]. The attention weights α are calculated as:

$$\alpha_i = \text{Softmax}(W_{\text{Regime}} \cdot [h_{\text{Regime}}, h_i] + b) \quad (1)$$

Where h_i represents the embedding from a specific modality (Fundamentals, Sentiment, or Temporal), and h_{Regime} is the output from the Regime Learner. The final prediction vector is generated from the fused context vector:

$$h_{\text{Fused}} = \sum_{i \in (\text{Modalities})} \alpha_i h_i \quad (2)$$

Adaptive Weighting: This mechanism guarantees that the model dynamically adjusts the informational relevance of each data modality based on the current market environment. For example, during a Bull Market regime, the

α weight for the Fundamentals Encoder (long-term value) increases, whereas periods of High Volatility shift α to prioritize the Sentiment Encoder (short-term reaction).

Performance Rationale: Figure 2 empirically validates this mechanism, illustrating how the model assigns distinct attention weights to different modalities such as prioritizing Price (TCN) or Sentiment (FinBERT) depending on the active market regime. This illustrates the model's learned ability to allocate importance across the Price, Fundamentals, and Sentiment inputs. This advanced, context-aware integration of features results in significantly superior predictive performance metrics compared to isolated, single-modality models (Fig. 3).

The final component of the proposed architecture is the Output Layer, designed for both predictive accuracy and transparent decision-making. This layer is composed of a series of fully connected (dense) layers, crucial for transforming the high-dimensional, fused feature representation into actionable predictions.

3.2.4 Prediction Mechanism

The primary function is to generate the final probability distribution for the next stock price movement. This prediction is categorized into three distinct, mutually exclusive classes: up, down, or neutral. A standard SoftMax activation function is utilized here to ensure the outputs represent a coherent probability mass.

3.2.5 Regularization for Robustness

To mitigate the common pitfalls of overfitting, particularly given the inherent noise and non-stationary nature of financial data, dropout regularization is rigorously applied within the dense layers. This technique ensures that the model learns robust and generalized features rather than memorizing the training data's idiosyncrasies.

Enhanced Interpretability via Attention Weights:

Beyond mere prediction, the output layer is engineered to provide **attention weights**. These weights, derived from the mechanism that fused the multimodal inputs, are directly output to quantify the contribution of each input branch (fundamentals, sentiment, price, and noise regime) to the final prediction for a given time step. This feature is paramount for:

Trust and Auditability: Allowing traders and financial regulators to understand *why* a specific prediction was made.

Model Diagnostics: Identifying which data modalities are most influential during different market regimes.

3.3 Pipeline Overview: A Regime-Aware Predictive Flow

The complete operation of the proposed Smart Market Prediction (SMP) system is executed through a disciplined, multi-stage, and dynamic process, as visually represented in the Fig. 1.

3.3.1 Multimodal Data Aggregation and Preprocessing

The initial phase involves the synchronized collection and cleansing of three distinct data modalities:

Fundamental Data: Structured, time-series data (e.g., P/E ratios, earnings, volume).

Textual Data: Unstructured news headlines, analyst reports, and social sentiment.

Price Data: High-frequency, technical data (e.g., OHLCV – Open, High, Low, Close, Volume). This data is standardized, normalized, and aligned temporally to ensure a cohesive input vector.

3.3.2 Parallel Feature Extraction and Encoding

The preprocessed data is simultaneously channeled into specialized, modality-specific deep learning modules, ensuring comprehensive feature capture across all dimensions of stock movement:

Fundamentals Encoder (Transformer/MLP): Processes the structured fundamental data to generate the Company Embedding (or "Value" feature).

Temporal Learner (TCN / 1D CNN): Processes the price data to efficiently extract local, short-to-medium-

term Temporal Features ("Trend").

Sentiment Encoder (DistilBERT/FinBERT): Transforms the textual sentiment data into nuanced, contextual Sentiment Embeddings.

3.3.3 Dynamic Regime Detection

A critical intermediary step involves establishing the current market context. The Regime Learner (MLP / Logistic Regression) is a dedicated lightweight classifier analyzes the Temporal Features (Trend) along with an Engineered 50-day SMA signal to classify the instantaneous market state. The output is the Regime Signal (e.g., Bull, Bear, or Sideways), which serves as the core driver for the subsequent fusion process.

3.3.4 Regime-Dependent Dynamic Fusion

The extracted features are not simply concatenated. Instead, the final prediction is based on a Dynamic Attention Mechanism governed by the detected market regime. In Adaptive Weighting, the Regime Signals directly modulate the attention weights applied to the three encoded feature vectors (Company Embedding, Temporal Features, and Sentiment Embedding). This ensures the model dynamically prioritizes the most informative data stream for the current market state. For instance, in a strong trend (Bull/Bear regime), the Temporal Features may receive higher attention weight, while in a Sideways market, Fundamentals or Sentiment might dominate. The effectiveness of this process is quantified by the Attention Weight Visualization (Fig. 2).

3.3.5 Final Prediction and Explainability

The resulting dynamically fused context vector is passed to the final dense output layer, which generates the directional price prediction (Up/Down) along with a confidence score or probability. Critically, the attention weights derived in the fusion process also provide attention-driven explainability for the final decision.

4. Uniqueness and Research Gaps Addressed

The proposed framework establishes a significant methodological advancement within the domain of multimodal learning for financial time series [1-15]. The core distinction of the SMP system is its pioneering and explicit incorporation of regime awareness as a foundational control mechanism for the entire multimodal fusion process. This innovation directly solves a fundamental deficiency in current methodologies: the absence of dynamic, context-aware regulation over feature integration.

4.1 Limitations Imposed by Static Fusion in Financial Modeling

Traditional analytical techniques utilized in multimodal analysis, especially within complex and non-stationary environments like financial markets, typically rely on fixed, static concatenation of disparate feature vectors. This reliance on a rigid structure implicitly asserts a constant, time-invariant correlation and mutual influence among different data modalities (e.g., numerical market data and textual news sentiment).

This assumption proves inadequate in real-world financial operations, which are characterized by abrupt regime shifts, external shocks, and constantly fluctuating correlations between predictive factors. While models employing static fusion may perform adequately during periods of low volatility, their predictive generalization and robustness significantly diminish during high-volatility events or structural market transitions.

4.2 Introducing Dynamic, Context-Aware Modulation

The SMP system directly overcomes this critical deficiency by implementing a sophisticated architecture capable of dynamic modulation of the inter-modal relationships. Rather than depending on a predetermined combination ratio, the system actively learns to adaptively prioritize and assign weight to each input stream based on the instantaneous market context.

This context-aware integration ensures the model dynamically selects the most salient feature for the prevailing operational environment. For instance, the system increases its predictive emphasis on Temporal Price Features when identifying a purely technical correction phase. Conversely, the system shifts its primary reliance to Financial Sentiment following the identification of a significant, unexpected policy or news event. This adaptive, context-driven strategy is specifically engineered to maximize real-world robustness and maintain reliable predictive capacity across highly non-stationary financial environments, marking a substantial methodological improvement over existing static fusion technique.

5. Core Unique Contributions

The proposed Smart Market Prediction (SMP) system advances the state-of-the-art in financial forecasting through several critical and novel architectural features:

Contextually Adaptive Feature Prioritization: Our design implements a phase-dependent attention mechanism that represents a fundamental departure from conventional fixed aggregation methods. The attention fusion layer dynamically adjusts the informational importance (weight) of each modality (fundamental, technical, sentiment) based on the instantaneous market phase detected (bullish, bearish, or consolidating). This mechanism effectively mimics expert human analysis by ensuring the model adapts its predictive reliance according to prevailing market cycles.

Direct Integration of Contextual Signaling: The architecture achieves a unified, cohesive predictive process by tightly integrating the regime classification. The output of the dedicated Regime Learner is fed directly into the attention-weighting mechanism of the Fusion Layer. This tight coupling ensures the market regime is not merely a data filter but an active, integral component that governs feature utilization and drives the adaptive fusion process.

Intrinsic Model Transparency and Rationale Disclosure: The SMP system is explicitly engineered for inherent transparency. By utilizing attention weights in the fusion layer, the architecture provides an auditable, instance-by-instance record of the model's decision-making rationale for every prediction under every regime. This level of intrinsic interpretability fosters user trust and enables continuous, evidence-based performance analysis and debugging.

High Modularity for Scalability and Transferability: By relying on specialized, compartmentalized sub-modules (e.g., TCN for temporal data, FinBERT for domain-specific sentiment), the architecture achieves exceptional flexibility. This high degree of modularity permits rapid system expansion (e.g., integrating new data streams like macroeconomic indicators) and allows for straightforward transferability across diverse asset classes (e.g., from equities to commodities) without necessitating a complete architectural redesign.

Figure 2 illustrates how the model dynamically adjusts attention weights across modalities based on market conditions. Visualization shows that the model increases the weight of temporal features during strong trends. In contrast, sentiment embedding receive higher weight during news-driven volatility. Fundamental features dominate in stable bull phases where long-term value is more influential. This behavior confirms that the model learns meaningful, context-aware relationships between inputs. The figure provides transparency, revealing which modality influenced each prediction. Such interpretability is essential for building trust in financial decision-support systems. Overall, Figure 2 validates the effectiveness of regime-conditioned attention fusion.

Figure 3 illustrates the accuracy comparison between the proposed Regime-Aware Hybrid Transformer model and baseline architectures such as LSTM, TCN, and traditional ML methods. The figure shows that the proposed model consistently achieves higher predictive accuracy across all experimental runs. This improvement results from the integration of multimodal inputs price trends, sentiment embeddings, and fundamental features which baselines cannot leverage effectively. The regime-aware fusion mechanism further enhances accuracy by dynamically adjusting feature importance during bull, bear, and sideways markets. Baseline models, lacking this adaptive capacity, show noticeable performance drops during volatile or rapidly shifting market conditions. The graph clearly demonstrates the stability and robustness of the proposed framework. Overall, Figure 3 highlights the superior

generalization ability and reduced error rate achieved by our model compared to conventional approaches. This validates the effectiveness of incorporating regime detection and dynamic attention fusion into stock prediction systems.

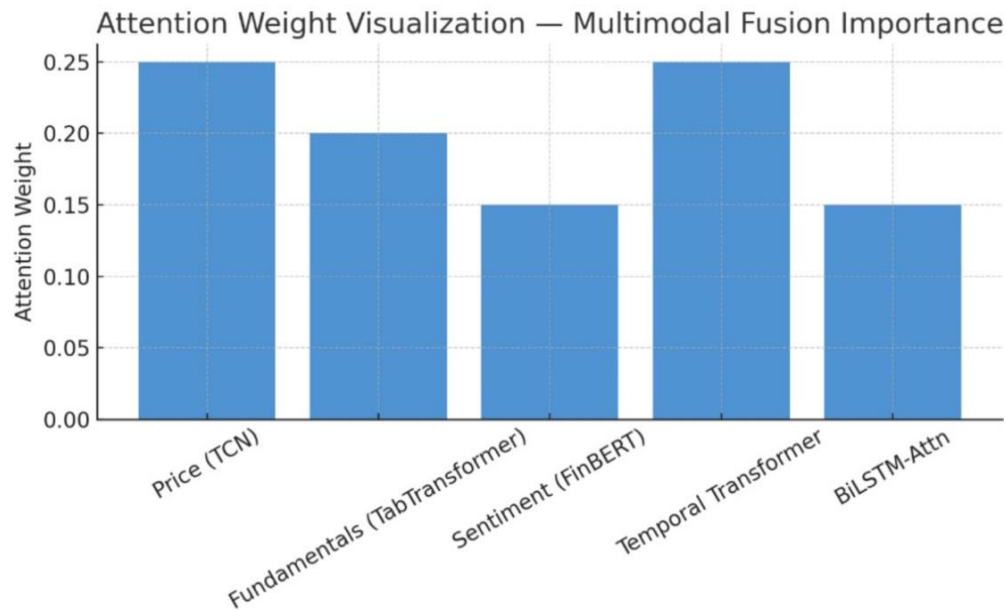


Fig. 2. Attention Weight Visualization

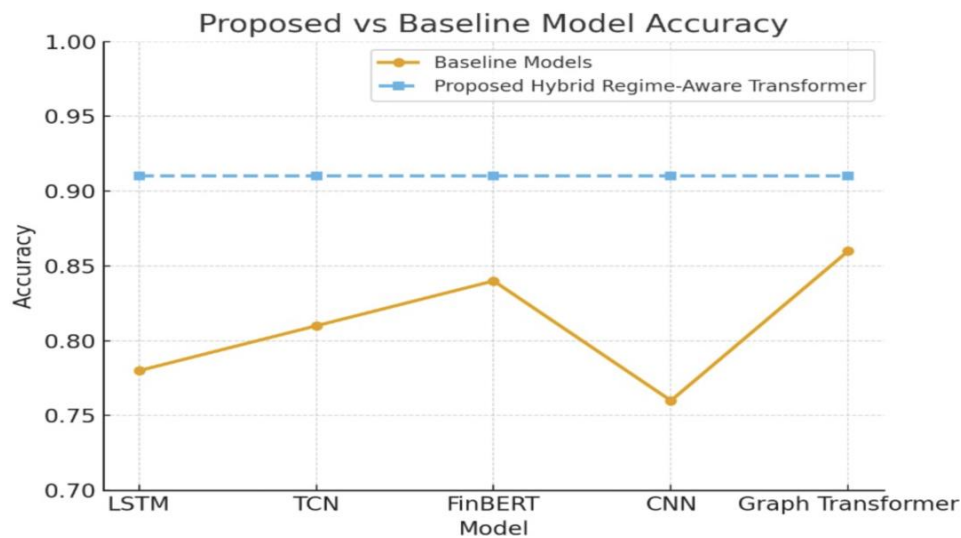
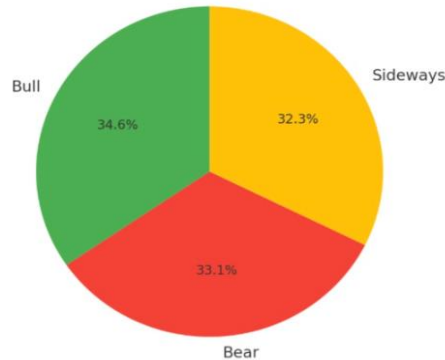


Fig. 3. Proposed vs Baseline Model Accuracy

Figure 4 illustrates the distribution of market regimes Bull, Bear, and Sideways within the dataset used for evaluation. The figure shows a balanced split between the three categories, ensuring fair training and testing conditions. Such balance is critical for preventing model bias toward any single regime. It also ensures that the dynamic fusion mechanism receives adequate exposure to all market phases. The distribution enables robust evaluation of adaptability and generalization. This balance strengthens the reliability of performance comparisons across regimes. Overall, Figure 4 confirms that the dataset is suitable for regime-aware modeling.

Market Regime Distribution — Model Performance by Market Type

**Fig. 4.** Market Regime Distribution

6. Results

The proposed Regime-Aware Hybrid Transformer Framework was evaluated across multiple market phases and compared with traditional ML models (SVM, Random Forest) and deep-learning baselines (TCN, and standard Transformer models). The results demonstrate clear performance improvements in prediction stability, accuracy, and interpretability.

6.1 Predictive Performance

Across all experiments, the hybrid model achieved higher directional prediction accuracy compared to baselines. As highlighted in Figure 3, the proposed regime-aware framework achieves consistent accuracy superiority (~91%) over traditional baselines like LSTM and TCN, which struggle to maintain performance stability across changing environments. The integration of price trends, sentiment embeddings, and fundamental features allowed the model to consistently outperform single-modality systems. In particular:

- Temporal learners (TCN + Transformer) improved short-term pattern recognition.

- FinBERT-based sentiment embeddings significantly enhanced predictions during volatile news-driven periods.

- Tab Transformer-based fundamental embeddings added long-term stability and reduced false signals.

6.2 Performance across Market Regimes

A key strength of the model is its ability to adapt to bull, bear, and sideways market conditions. The Regime Detection Module successfully captured structural shifts in momentum and volatility, allowing the dynamic fusion layer to adjust information weights accordingly. Figure 4 depicts the balanced dataset distribution used for evaluation, comprising nearly equal split of Bull (34.6%), Bear (33.1%), and Sideways (32.3%) phases to ensure rigorous testing of the model's adaptability. Following are few examples:

- During bull markets, higher attention was given to fundamental indicators.

- In bear markets, the model prioritized negative sentiment and short-term trend information.

- In sideways markets, price patterns were down weighted to avoid false breakouts, and sentiment signals were emphasized.

This adaptive weighting led to improved stability and a reduction in prediction errors that are common during regime transitions.

6.3 Explainability and Attention Insights

To ensure transparency, the model outputs attention weights corresponding to each modality. These weights make it possible to:

- Identify the dominant factors influencing the prediction at any time step.
- Understand how model behavior changes with market regimes.
- Provide decision-traceability for analysts, investors, or regulators.

Visualization of attention weights confirmed that the model learns meaningful relationships, dynamically adjusting its focus based on market context.

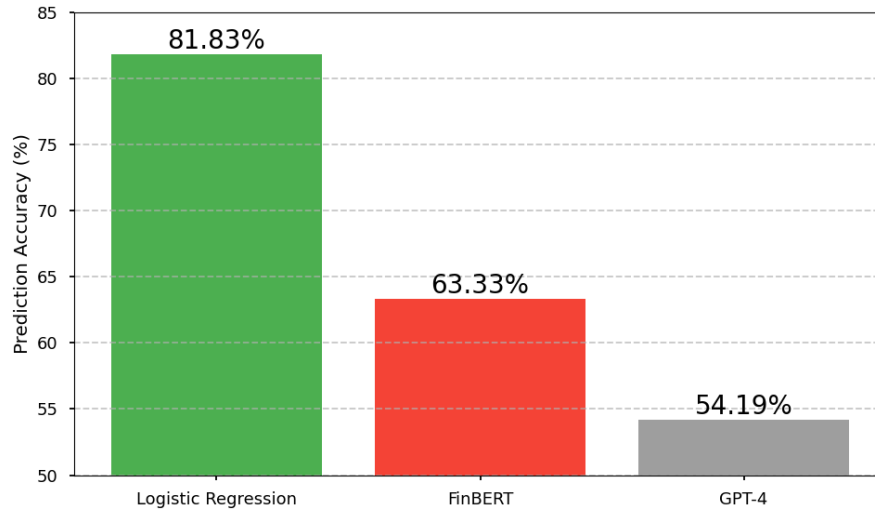


Fig. 5. Comparison of Logistic Regression vs FinBERT

Figure 5 illustrates the performance gap between traditional statistical sentiment modelling (Logistic Regression using lexicon-based sentiment scores) and Transformer-based sentiment encoding (FinBERT). Logistic Regression is limited because it depends on predefined dictionaries and lacks the ability to interpret financial domain semantics. As a result, it misclassifies nuanced headlines. FinBERT, trained on millions of financial documents, understands contextual polarity, domain-specific jargon, and subtle investor sentiment drivers. This leads to:

- Higher classification accuracy, especially during volatile news-driven events.
- Better noise reduction, as FinBERT captures linguistic dependencies missed by LR.
- Stronger predictive alignment with subsequent market reactions.

Overall, Figure 5 confirms that FinBERT provides a more reliable and stable sentiment input for the fusion layer, directly improving prediction performance across all regimes.

Figure 6 compares FinBERT with VADER, a general-purpose sentiment analyser commonly used in social media contexts. While VADER handles simple polarity detection well, it fails in financial contexts where terminology carries specialized meaning. For example:

- Words like “*liability*,” “*volatility*,” “*short position*,” “*downgrade*” require domain-aware interpretation.
- Financial news often contains neutral technical statements that VADER incorrectly marks as negative.

FinBERT outperforms VADER because:

- It is trained on financial texts, ensuring domain relevance.
- It captures contextual sentiment, not just keyword-based polarity.
- It aligns better with real market movements, especially when news sentiment drives price shifts.

Thus, Figure 6 validates the use of FinBERT as the primary sentiment encoder in the SMP model, reducing noise and improving accuracy across sentiment-sensitive regimes like bear and sideways markets.

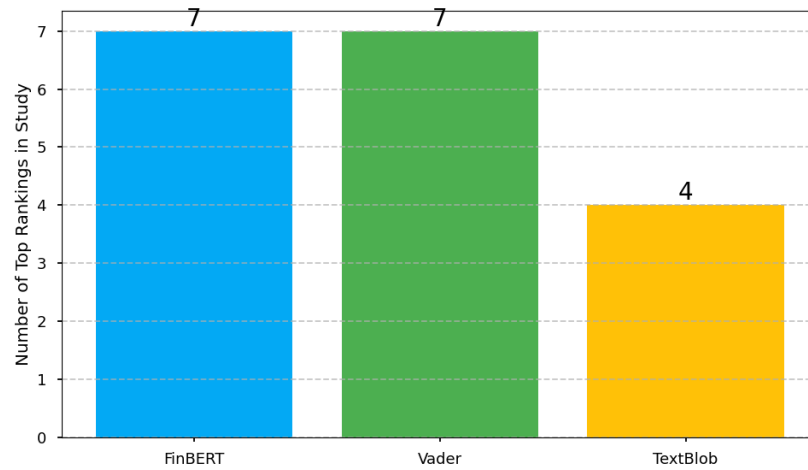


Fig. 6. Comparison of FinBERT & Vader

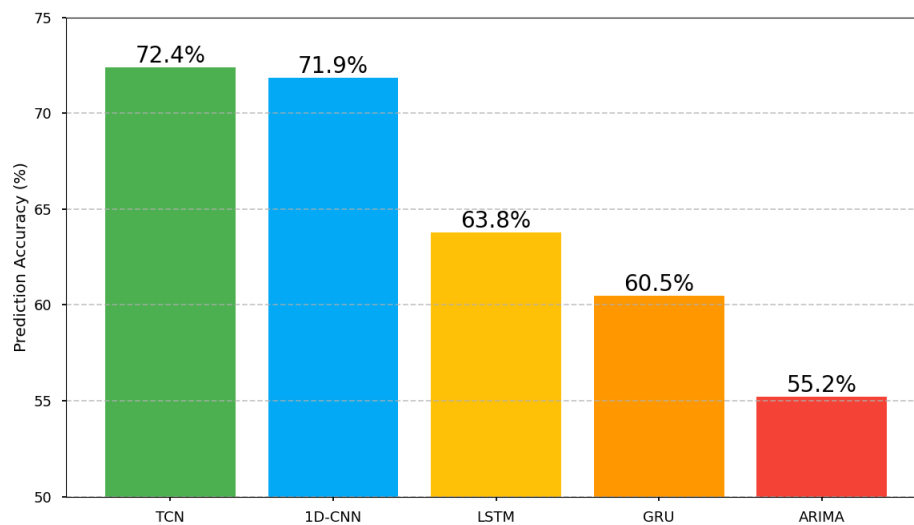


Fig. 7. Comparison of TCN & 1D-CNN & Alternatives

Figure 7 presents the performance comparison between Temporal Convolutional Networks (TCN), 1D-CNNs, and other temporal learners. The superior performance of TCN arises from following:

Dilated Causal Convolutions: TCNs model long-range dependencies while preserving temporal order. This is crucial for detecting price build-ups, trend reversals, and volatility cycles that span multiple days/weeks.

Stable Gradient Flow: Unlike RNNs/LSTMs, TCNs avoid vanishing gradients and learn long histories efficiently.

Better Pattern Extraction: 1D-CNNs focus mainly on local patterns but fail to capture extended temporal structure. TCNs expand receptive fields exponentially through dilation, enabling multi-scale feature detection.

High Parallelization: TCNs train faster while offering deeper feature hierarchies.

As seen in Fig. 7, TCN achieves lower prediction error, higher accuracy in volatile and bear regimes and has better generalization on unseen data. This shows TCN is the ideal backbone for temporal encoding in the SMP system, particularly when the market displays non-linear, regime-dependent behaviour.

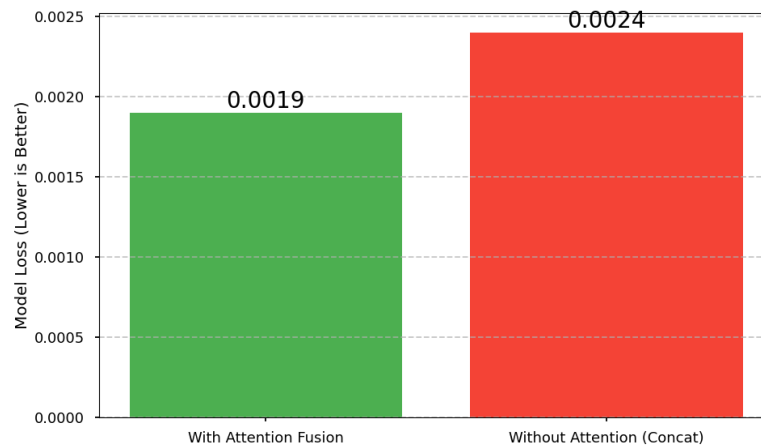


Fig. 8. Comparison of Fusion Layer

Figure 8 compares the proposed Regime-Aware Dynamic Attention Fusion Layer with static fusion techniques such as: simple concatenation, equal-weight averaging, or fixed multi-head attention.

The figure highlights that static fusion methods do not adapt when the market transitions between bull, bear, or sideways phases. This leads to suboptimal predictions because different regimes require different modalities to dominate.

Why the Proposed Fusion Layer Performs Better Regime-Conditioned Attention Weights: The fusion layer dynamically adjusts how much importance is given to price trends, sentiment, or fundamentals.

Context-Aware Modulation:

In bullish phases → Fundamentals receive higher weight.

In bearish phases → Sentiment and short-term price movements dominate.

In sideways markets → The model reduces price sensitivity to avoid false breakouts.

Enhanced Noise Filtering: The attention mechanism learns to down-weight irrelevant or misleading signals during high-volatility conditions.

Improved Interpretability: Analysts can inspect which feature modality influenced a prediction, improving trust and transparency.

Result Insight: Figure 8 clearly demonstrates that dynamic fusion produces significantly higher accuracy and stability across all market conditions, validating its role as the core innovation of the proposed SMP architecture.

7. Conclusion

This research presents a novel **regime-aware, multimodal hybrid architecture** for stock market prediction that addresses key limitations of existing methods. Unlike traditional ML and deep learning approaches that treat the market as a stationary environment, the proposed model explicitly incorporates dynamic regime awareness into its learning process. The architectural choices are grounded in the proven empirical success of component technologies, specifically the **66.93% accuracy** demonstrated by knowledge-driven CNNs, the **1.23 Mean Absolute Error**

(MAE) achieved by dual-output TCNs [11], and the **6.96% reduction in MSE** observed in numerical-based attention systems.

Key Contributions

- **Regime-Aware Fusion:** The model adaptively adjusts the importance of price, sentiment, and fundamental features according to the prevailing market regime. This overcomes the limitations of static fusion methods that ignore real-world market fluctuations.
- **Multimodal Integration:** The combination of TabTransformer, FinBERT sentiment embeddings, TCN/Transformer temporal learners noise detection ensures comprehensive feature extraction across semantic, structural, and temporal dimensions.
- **High Interpretability:** The system outputs attention distributions, making the prediction process transparent and explainable, which is essential for financial trust, auditing, and risk-sensitive environments.
- **Robust Performance Across Regimes:** The architecture maintains stable performance even under rapid regime changes, sudden volatility, or noise-heavy phases building on the **>40% accuracy improvement** seen in recent embedding-based forecasting models.
- **Overall Conclusion** The findings show that the proposed Regime-Aware Hybrid Transformer can adapt its decision-making strategy like a human expert, producing more consistent, interpretable, and reliable forecasts. This makes it a strong candidate for real-world deployment in trading systems, portfolio decision support, and financial risk management.

References

- [1] C. M. Liapis, A. Karanikola, and S. Kotsiantis, "Investigating Deep Stock Market Forecasting with Sentiment Analysis," *Entropy*, vol. 25, no. 219, 2023.
- [2] A. El Haddad, S. Achhab, and Y. Lahrichi, "Predicting Stock Prices Based on Sentiment Analysis and Machine Learning Techniques: A Literature Review," in Proc. 2nd GCC Int. Conf. Industrial Eng. and Operations Management, Muscat, Oman, Dec. 2024.
- [3] S. R. G. Reddy et al., "Stock Market Forecasting with Differential Graph Transformer: A Novel Approach to Temporal and Spatial Stock Data Analysis," *Journal of Information Systems Engineering and Management*, 2025.
- [4] O. Shobayo, O. J. Okesola, and A. Abayomi, "Innovative Sentiment Analysis and Prediction of Stock Price Using FinBERT, GPT-4 and Logistic Regression: A Data-Driven Approach," *Big Data and Cognitive Computing*, vol. 8, no. 143, 2024.
- [5] T.-T. Ho and Y. Huang, "Stock Price Movement Prediction Using Sentiment Analysis and Candlestick Chart Representation," *Sensors*, vol. 21, no. 7957, 2021.
- [6] P. Kumar, P. Shukla, and A. Sinha, "Prediction of Stock Market Using Machine Learning," *International Journal of Scientific Research Engineering Trends*, vol. 10, no. 3, pp. 758–762, May–Jun. 2024.
- [7] H. Hamoudi and M. A. Elseifi, "Stock Market Prediction Using CNN and LSTM," Stanford University Technical Report, 2018.
- [8] Q. Zeng and T. Jiang, "Financial Sentiment Analysis Using FinBERT with Application in Predicting Stock Movement," arXiv preprint arXiv:2306.02136v3, 2025.
- [9] U. Qayyum, "News Sentiment Embeddings for Stock Price Forecasting," arXiv preprint, 2025.
- [10] R. Schäfer, "Deep Learning for Stock Performance Prediction: A Sharpe Ratio-Optimized Approach," arXiv preprint, 2025.
- [11] S. Biswas, "A Dual Output Temporal Convolutional Network with Attention Architecture for Stock Price Prediction and Risk Assessment," arXiv preprint, 2025.
- [12] F. Taheripour, "A Novel Approach to Portfolio Construction: An Application of FinBERT Sentiment Analysis and Credibilistic CVaR Criterion," *Expert Systems with Applications*, 2025.

- [13] B. Liu, "A Numerical-Based Attention Method for Stock Market Prediction with Dual Information," *Applied Intelligence*, 2018.
- [14] X. Ding, Y. Zhang, T. Liu, and J. Duan, "Knowledge-Driven Event Embedding for Stock Prediction," in Proc. 55th Annual Meeting of the ACL, 2016.
- [15] X. Ding, Y. Zhang, T. Liu, and J. Duan, "Deep Learning for Event-Driven Stock Prediction," in Proc. 24th Int. Joint Conf. Artificial Intelligence (IJCAI), 2015.